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ORIGINAL SCIENTIFIC REPORT

Prophylactic Central Neck Dissection for Papillary Thyroid Carcinoma with Clinically Uninvolved Central Neck Lymph Nodes: A Systematic Review and Meta-analysis

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Abstract

Background Central neck dissection and total thyroidectomy are standard treatments for patients with papillary thyroid carcinoma (PTC) with clinically involved central nodes. However, prophylactic central neck dissection (pCND) in patients with clinically uninvolved cN0 has been beneficial in some studies but ineffective in others. We conducted a meta-analysis to evaluate the efficacy and safety of pCND in patients with central neck lymph nodes cN0 PTC.

Methods The PubMed, EMBASE, Cochrane Library, and Scopus databases and the ClinicalTrials.gov registry were electronically searched for studies published until September 2017. The meta-analysis was conducted to calculate the pooled effect size by using random-effects model. Treatment efficacies were measured by determining locoregional recurrence (LRR). Secondary outcomes included transient recurrent laryngeal nerve (RLN) injury, permanent RLN injury, transient hypocalcemia, and permanent hypocalcemia.

Results Twenty-three retrospective and prospective cohort studies involving 18,376 patients were reviewed. Patients who underwent pCND had significantly lower LRR (odds ratio [OR] 0.65; 95% confidence interval [CI] 0.48–0.88) but significantly higher incidence rates of transient RLN injury (OR 2.03; 95% CI 1.32–3.13), transient hypocalcemia (OR 2.23; 95% CI 1.84–2.70), and permanent hypocalcemia (OR 2.22; 95% CI 1.58–3.13) than that of no pCND group.

Conclusion Compared with no pCND, pCND significantly reduces LRR but is accompanied by numerous adverse effects. The clinical decision should be made after the shared decision-making process of clinicians and patients.

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Introduction

Papillary thyroid carcinoma (PTC) is the most common type of differentiated thyroid carcinoma, accounting for approximately 90% of new cases of thyroid cancer [1]. Although the 10-year survival rate of patients with PTC exceeds 90%, recurrence has been reported in up to 30% of patients [2, 3]. Total thyroidectomy (TT) is the standard surgical procedure for patients with PTC. Other surgical strategies such as hemithyroidectomy and thyroid lobectomy are also viable.

Regional lymph node metastasis is detected in 40–90% of recurrent cases [4, 5] and is an independent risk factor for locoregional recurrence (LRR) [3, 6] and mortality in patients with PTC [7]. However, lymph node metastases are often microscopic and difficult to detect preoperatively [8]. Certain studies have indicated that prophylactic central neck dissection (pCND) prevents future recurrence [4, 9], whereas other studies have found no significant difference in LRR and observed increased rates of hypocalcemia and recurrent laryngeal nerve injury [10, 11]. The American Thyroid Association Management guidelines recommend the use of central neck dissection for patients with clinically involved central nodes; however, the use of pCND in patients with clinically uninvolved central neck lymph nodes (cN0) is under heavy debate [12]. Nevertheless, Huang et al. indicated that the recommendation of pCND varies in different guidelines [13].

Conducting randomized controlled trials to evaluate the efficacy of pCND is difficult. Although many retrospective and prospective studies have been conducted, their results are variable. Two meta-analyses including studies published before 2015 have investigated the necessity of pCND [14, 15]. However, both meta-analyses have included studies conducted on patients who had already shown evidence of node metastases preoperatively. This potentially skews the outcome of the analyses because patient populations with different baselines should not be compared with each other. In addition, several large-scale studies have been published in recent years. Therefore, we conducted a systematic review and meta-analysis of the evidence available up to date to investigate the efficacy and necessity of pCND in patients with cN0 PTC.

Materials and methods

Inclusion and exclusion criteria

In this review, we included clinical studies evaluating the outcome of pCND in patients with PTC. Studies were also required to clearly report the patient inclusion and exclusion criteria, surgical strategy, and tumor stage, as well as the definition and evaluation of prognostic outcomes. We excluded studies that met at least one of the following criteria: (1) patients with clinical evidence of nodal metastases, (2) patients with a previous history of thyroidectomy, (3) patients with mixed-type PTC, or (4) patient cohorts reported in duplicates.

Search strategy and study selection

The PubMed, EMBASE, Scopus, and Cochrane databases were electronically searched for relevant studies published until September 2017. The following medical subject headings were used: *(((papillary thyroid carcinoma) AND thyroidectomy)) AND ((neck dissection) OR lymph node dissection)*. The "similar articles" option in PubMed was used for broadening the search, and all retrieved abstracts, studies, and citations were reviewed. In addition, we identified other studies by using the reference sections of relevant papers and through correspondence with subject experts. Finally, unpublished studies were collected from the ClinicalTrials.gov registry (http://clinicaltrials.gov/). No language restrictions were applied. The systematic review described herein has been accepted by PROSPERO, an online international prospective register of systematic reviews, curated by the National Institute for Health Research (CRD42017069976).

Data extraction

Baseline and outcome data were independently abstracted by two reviewers (LC and YHW), and the study design, population characteristics, inclusion and exclusion criteria, surgical strategy, complications, and posttreatment parameters were extracted. Decisions individually recorded by the reviewers were compared, and disagreements were resolved by a third reviewer (KWT). The authors of the collected studies were contacted for additional information.

Methodological quality appraisal

Two reviewers (LC and YHW) independently assessed the methodological quality of each study by using the Risk of Bias In Non-randomized Studies of Interventions assessment tool recommended by the Cochrane Collaboration [16]. Several domains of biases were assessed, including the control of confounding variables, selection of participants, classification of interventions, deviations from intended interventions, completeness of data, measurement of outcomes, and selective reporting.

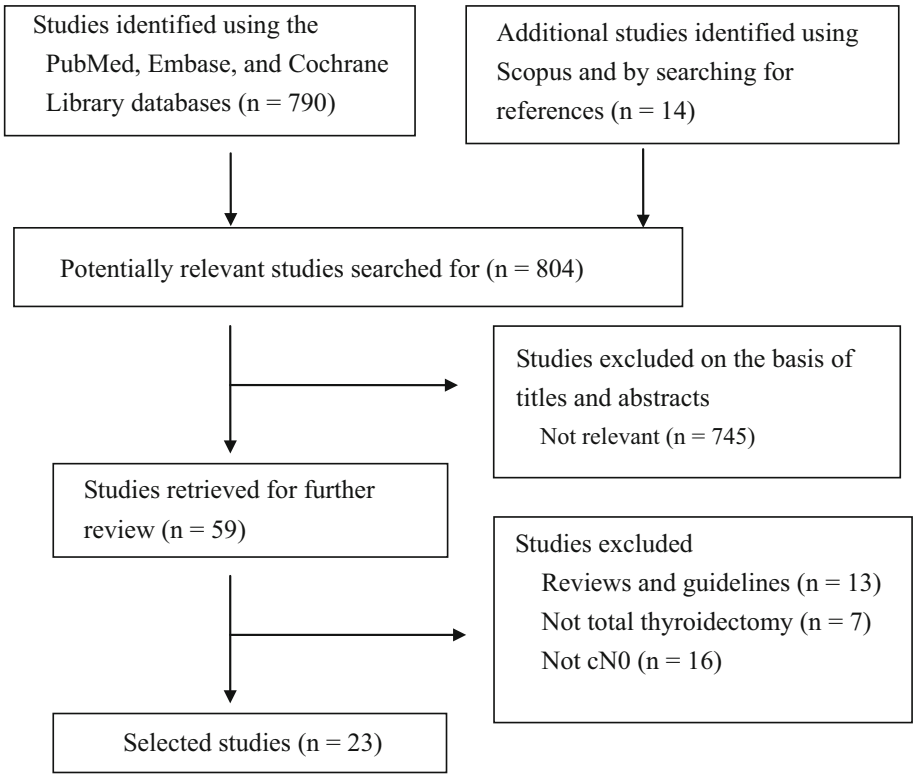

Fig. 1 Flowchart of study inclusion criteria

Outcomes

The primary outcome was LRR. Secondary outcomes included transient recurrent laryngeal nerve (RLN) injury, permanent RLN injury, transient hypocalcemia, and permanent hypocalcemia. Of 17 studies that investigated RLN injury, 3 studies [9, 17, 18] considered RLN injury to be permanent when the symptoms were observed for a period of 12 months after operation, whereas the remaining 14 defined permanent RLN injury as the occurrence of symptoms for more than 6 months after initial surgery. Of 19 studies that measured hypocalcemia, 3 studies [9, 17, 18] defined the duration of "transient" as 12 months or shorter, whereas the remaining defined it as within 6 months.

Statistical analyses

Data were entered into and analyzed using Review Manager, version 5.3 (The Cochrane Collaboration, Oxford, England). The meta-analysis was performed in accordance with PRISMA guidelines [19]. Standard deviations were estimated from the provided confidence interval (CI) limits or the standard error. Dichotomous outcomes were analyzed using odds ratio (OR) as the summary statistic. The precision levels of effect sizes were reported as 95% CIs. A pooled estimate of the OR was computed using the DerSimonian and Laird random-effects model [20].

The statistical heterogeneity and inconsistency of treatment effects across the studies were evaluated using Cochrane's Q test and I^2 statistics, respectively. Statistical significance was set at $p < 0.10$ for Cochran's Q test. Moreover, statistical heterogeneity across the studies was assessed using I^2 statistics, which quantified the proportion of the total outcome variability across the studies.

Results

Study characteristics

Figure 1 presents a flowchart describing the study screening and selection procedure. The initial search strategy yielded 804 studies, of which 745 were ineligible based on the criteria used for screening titles and abstracts. Therefore, the full texts of the remaining 59 studies were retrieved. However, most of these studies were excluded from our final review because of the following reasons: 13 were reviews or guidelines, 7 did not perform TT, and 16 included patients who did not have cN0. Finally, 23 studies were included in the current review [4, 9–11, 17, 18, 21–36].

The 23 studies included in this review were published between 2007 and 2017 and had sample sizes ranging from 81 to 11,569 patients [4, 9–11, 17, 18, 21–36]. All studies enrolled patients who had PTC but no clinical evidence of nodal involvement. Patients underwent thyroidectomy with or without pCND. The majority of the enrolled patients were female, except those of two studies [11, 32]. The mean tumor size ranged from 0.5 to 2.3 cm. All studies had enrolled patients who underwent total thyroidectomy, and only one study enrolled patients who underwent hemithyroidectomy [25]. More patients in pCND group received radioactive ^{131}I ablation (RAI) compared with no pCND group. The baseline characteristics of the treatment groups of the included studies are shown in Table 1.

Table 2 shows a summary of the methodological quality of the included studies. At preintervention, two studies [17, 21] had a moderate risk of bias due to confounding. Fourteen studies [11, 17, 21–29, 31, 32] had an unclear risk of selection bias because the decision of performing pCND was at the discretion of the surgeon. All studies were rated as having a low risk of bias in the classification of interventions during the intervention. Furthermore, all studies had a low risk of bias of deviation from intended interventions and a bias in the measurement of outcomes after the intervention. However, one study [18] was rated as having a serious risk of bias because of missing data and a bias in the selection of reported results because only a fraction of the total participants was further analyzed. Overall, 10 studies [4, 5, 9, 10, 21, 30, 33–36] were rated as having a low risk of bias, 12 studies [11, 17, 22–29, 31, 32] were rated as having a moderate risk of bias, and 1 study [18] was rated as having a serious risk of bias.

LRR investigation

Twenty studies investigated LRR (Fig. 2) [4, 5, 9–11, 17, 18, 21, 22, 24–29, 31, 33–36]. Recurrence was defined as an increasing thyroglobulin or antithyroglobulin antibody level [4, 9, 10, 18, 26, 29, 34]; macroscopic diseases at clinical examination [24, 28]; positive imaging on ultrasound, computerized axial tomography or radioiodine whole-body scan [4, 9, 24, 26, 28, 29, 32, 34, 35]; or reoperation with confirmed pathology [4, 35]. LRR occurred in 280 (2.52%) of 11,098 patients in the pCND group and in 254 (4.59%) of 5583 patients in the no pCND group. Disease recurrence was significantly lower in patients who underwent pCND (OR 0.65; 95% CI 0.48–0.88; Fig. 2).

Transient RLN injury

Seventeen studies examined transient RLN injury (Fig. 3) [4, 5, 9–11, 17, 18, 22, 23, 25, 27–32, 35]. Three studies [9, 17, 18] defined the duration of "transient" as within 12 months, whereas the remaining studies defined it as within 6 months. The incidence of transient RLN injury was significantly higher in the pCND group (157/4449, 3.53%) than in the no pCND group (94/4725, 1.99%; OR 2.03; 95% CI 1.32–3.13; Fig. 3).

Permanent RLN injury

Eighteen studies examined permanent RLN injury (Fig. 4) [4, 5, 9–11, 22–32, 34, 35]. Three studies [9, 17, 18] defined the term "permanent" as a symptom duration of more than 12 months, whereas the remaining studies defined it as a period of more than 6 months. Permanent RLN injury developed in 42 (0.90%) of 4651 patients in the pCND group and in 36 (0.75%) of 4826 patients in the no pCND group. No significant difference was observed in the incidence of permanent RLN injury between the two groups (OR 1.31; 95% CI 0.75–2.27; Fig. 4).

Transient hypocalcemia

Nineteen studies examined transient hypocalcemia (Fig. 5) [4, 5, 9–11, 17, 18, 22–32, 35]. Three studies [9, 17, 18] defined the duration of "transient" as within 12 months, whereas the remaining studies defined it as within 6 months. Transient hypocalcemia was observed in 1209 (25.75%) of 4696 patients who underwent pCND and in 726 (14.83%) of 4894 patients who did not undergo pCND. The incidence of transient hypocalcemia was significantly higher in patients who underwent pCND (OR 2.23; 95% CI 1.84–2.70; Fig. 5).

Permanent hypocalcemia

Nineteen studies assessed permanent hypocalcemia (Fig. 6) [4, 5, 9–11, 17, 22–32, 34, 35]. Two studies [9, 17] defined the duration of "permanent" as more than 12 months, whereas the remaining studies defined it as more than 6 months. The incidence of permanent hypocalcemia was significantly higher in patients who underwent pCND (193/4699, 4.11%) than in those who did not undergo pCND (95/4879, 1.95%; OR 2.22; 95% CI 1.58–3.13; Fig. 6).

Other measurements

Overall survival was reported in one study, and disease-free survival was reported in two studies. Barczynski et al. [9] reported a significantly higher 10-year overall survival rate for the TT + pCND group (96.2%) than for the TT only group (91.5%). Costa et al. [21] found no significant difference in disease-free survival between the pCND

Table 1 Characteristics of included studies

Study	Inclusion criteria	Number of patients (% male)	Age, year, mean ± SD (patient number)	Tumor size, cm, mean ± SD	Radioactive ^{131}I ablation, patient number (%)	Length of follow-up, month, mean ± SD	Intervention
Barczyński et al. [9]	PTC, cN0, no distant metastases	P: 358 (21) C: 282 (21.3)	P: <45: 43.6 (156) ≥45: 56.4 (202) C: <45: 43.3 (122) ≥45: 56.7 (160)	N/P	P: 231 (64.5) C: 79 (28)	P: 120 (12–132)[a] C: 120 (12–134)[a]	BP: TT + bpCND C: TT
Calò et al. [10]	PTC, cN0	P: 65 (18.5) C: 220 (21.8)	P: 46.36 ± 14.73 C: 53.07 ± 13.9	P: 1.7 ± 0.9 C: 1.6 ± 1.2	P: 65 (100) C: 194 (88.2)	100 (38–137)[a]	P: TT + pCND C: TT
Calò et al. [18]	PTC, cN0	IP: 30 (13.3) BP: 30 (20.0) C: 103 (25.2)	IP: 45.8 ± 10.5 BP: 39.7 ± 10.5 C: 45.4 ± 12.2	IP: 1.35 ± 0.75 BP: 1.24 ± 0.49 C: 1.42 ± 0.9	IP: 19 (63.3) BP: 26 (86.7) C: 59 (57.3)	IP: 71.1 ± 9.8 BP: 72.5 ± 8.4 C: 46.1 ± 27.8	IP: TT + ipCND BP: TT + bpCND C: TT
Choi et al. [17]	PTC, tumor ≤1 cm, cN0	P: 48 (12.5) C: 53 (20.8)	P: 52 ± 12.0 C: 48 ± 9.9	P: 0.68 ± 0.21 C: 0.73 ± 0.23	P: 48 (100) C: 53 (100)	24.4 (13–55)[b]	P: TT + bpCND C:TT
Costa et al. [21]	PTC, cN0	P: 126 (20) C: 118 (21)	P: <45: 49 (62) ≥45: 51 (64) C: <45: 27 (32) ≥45: 73 (86)	P: 1.7 (1–6.0)[b] C: 1.5 (0.1–10)[b]	P: 87 (69) C: 62 (53)	P: 47 (0–159)[b] C: 64 (0–144)[b]	P: TT + pCND C: TT
Dobrinja et al. [22]	PTC, cN0	P: 74 (16.2) C: 112 (25.9)	P: 53 ± 16.3 C: 57 ± 14.6	P: 1.3 (0.9–1.8)[b] C: 1.1 (0.7–1.6)[b]	P: 43 (58.1) C: 55 (49.1)	P: 37 (22–66)[b] C: 76 (35–112)[b]	P: TT + pCND C: TT
Giordano et al. [23]	PTC, cN0	IP: 385 BP: 308 C: 394	N/P	N/P	N/P	N/P	IP: TT + ipCND BP: TT + bpCND C: TT
Hughes et al. [24]	PTC, tumor >1 cm, cN0, no distant metastases	P: 78 (21.8) C: 65 (24.6)	P: 46.8[d] C: 41.2[d]	P: 1.9[d] C: 2.0[d]	P: 72 (92) C: 52 (86)	P: 19.1[d] C: 27.5[d]	BP: TT + bpCND C: TT
Hyun et al. [25]	PTC, tumor ≤1 cm, cN0	P: 65 (14) C: 87 (23)	P: 46 C: 48	N/P	N/P	51.3 ± 25.9	P: HT + upCND C: HT
Korkmaz et al. [26]	PTC, cN0	P: 162 (11.1) C: 140 (14.2)	P: 42.3 ± 12.9 C: 49.6 ± 12.0	P: 1.21 ± 0.96 C: 1.32 ± 1.33	P: 139 (85.8) C: 104 (74.8)	P: 30.7 ± 17.1 C: 30.1 ± 16.7	P: TT + pCND C: TT
Kim et al. [27]	PTC, age >18 years, cN0	P: 8735 (19.9) C: 2834 (19.0)	P: <45: 42.2 (3682) ≥45: 57.8 (5053) C: <45: 44.9 (1272) ≥45: 55.1 (1562)	P: 0.9 ± 0.6 C: 0.8 ± 0.7	P: 5307 (60.8) C: 1146 (40.4)	62.6 (6.0–216.6)[a]	P: TT + bpCND C: TT

Table 1 continued

Study	Inclusion criteria	Number of patients (% male)	Age, year, mean ± SD (patient number)	Tumor size, cm, mean ± SD	Radioactive ^{131}I ablation, patient number (%)	Length of follow-up, month, mean ± SD	Intervention
Lang et al. [28]	PTC, cN0	P: 82 (22.0) C: 103 (21.4)	P: 52.0 (17.1–79.2)[b] C: 50.0 (26.2–90.8)[b]	P: 1.5 (0.2–9)[b] C: 1.0 (0.2–11)[b]	P: 62 (75.6) C: 63 (68.0)	P: 25.5 (9.7–77)[a] C: 27.1 (10.7–101)[a]	P: TT + upCND C: TT
Lee et al. [29]	PTC, cN0	P: 153 (15.4) C: 104 (19.6)	P: 51.6 ± 3.4 C: 52.3 ± 2.3	P: 1.6 ± 1.4 C: 1.7 ± 1.2	P: 112 (73.2) C: 74 (71.2)	P: 55.2 ± 11.3 C: 49.2 ± 11.9	P: TT + pCND C: TT
Moo et al. [4]	PTC, cN0	P: 45 (22.2) C: 36 (11.1)	P: 49.2 C: 45.7	P: 2.04 C: 1.42	P: 31 (68) C: 26 (72)	P: 37 C: 37	P: TT + bpCND C: TT
Palestini et al. [30]	PTC	P: 93 (29.0) C: 148 (20.9)	P: 44 (15–82) C: 52 (17–80)	N/P	N/P	N/P	P: TT + ipCND C: TT
Popadich et al. [5]	PTC, tumor ≥1 cm, age ≥16 y, cN0	P: 259 (20.1) C: 347 (23.3)	P: 44 C: 48	P: 2.27 C: 2.23	593 (98)	P: 32 C: 50	P: TT + pCND C: TT
Raffaelli et al. [31]	PTC, cN0	UP: 62 (17.7) BP: 62 (21.0) C: 62 (21.0)	UP: 44.0 ± 11.2 BP: 41.4 ± 11.9 C: 43.2 ± 11.3	UP: 1.2 ± 0.6 BP: 1.4 ± 0.8 C: 1.2 ± 0.6	UP: 42 (68) BP: 48 (77) C: 37 (60)	UP: 25.1 ± 8.5 BP: 24.9 ± 8.9 C: 22.5 ± 6.6	UP: TT + upCND BP: TT + bpCND C: TT
So et al. [32]	PTC, cN0	P: 119 (82.4) C: 113 (85.8)	P: 49.18 ± 9.37 C: 49.75 ± 10.50	P: 0.66 ± 0.21 C: 0.62 ± 0.20	P: 101 (84.9) C: 92 (81.4)	N/P	P: TT + bpCND C: TT
Sywak et al. [33]	PTC, tumor >1 cm, cN0	P: 56 (28.6) C: 391 (27.9)	P: 39.1 (35–43) C: 42.6 (37–47)	P: 2.0 (1.5–2.3)[c] C: 2.3 (1.8–2.9)[c]	P: 56 (100) C: 391 (100)	P: 24.5[d] C: 70[d]	P: TT + upCLND C: TT
Viola et al. [34]	PTC, cN0	P: 93 (26.9) C: 88 (23.9)	P: 45.7 ± 14.6 C: 43.5 ± 13.2	P: 1.6 ± 0.8 C: 1.6 ± 0.9	175 (96.7)	59.4 ± 7.1	P: TT + bpCND C: TT
Ywata de Carvalho et al. [11]	PTC, cN0	P: 102 (85.3) C: 478 (88.5)	P: 41.2 (13–75) C: 45.2 (17–84)	P: 1.5 (0.4–4) C: 1.0 (0.1–12.5)	P: 58 (56.9) C: 212 (44.4)	P: 80.2 C: 67.4	P: TT + pCND C: TT
Zhang et al. [35]	PTC, tumor <1 cm, cN0	P: 134 (19.4) C: 108 (25.0)	P: 48 C: 45	P: 0.7 C: 0.5	N/P	P: 61[d] C: 66[d]	P: TT + pCND C: TT

Table 1 continued

Study	Inclusion criteria	Number of patients (% male)	Age, year, mean ± SD (patient number)	Tumor size, cm, mean ± SD	Radioactive ^{131}I ablation, patient number (%)	Length of follow-up, month, mean ± SD	Intervention
Zuniga and Sanabria [36]	PTC, cN0	P: 136 (7.3) C: 130 (10.0)	P: 42.9 (14.1) C: 41.5 (13.3)	N/P	P: 79 (58.2) C: 55 (42.3)	82.8 ± 51.6	P: TT + pCND C: TT

BP bilateral prophylactic, *bpCND* bilateral prophylactic central neck dissection, *C* control, *cN0* clinically uninvolved lymph nodes, *HT* hemithyroidectomy, *IP* ipsilateral prophylactic, *ipCND* ipsilateral prophylactic central neck dissection, *LN* lymph node, *N/A* not provided, *P* prophylactic, *pCND* prophylactic central neck dissection, *PTC* papillary thyroid carcinoma, *RCT* randomized controlled trial, *TT* total thyroidectomy, *UP* unilateral prophylactic, *upCND* unilateral prophylactic central neck dissection

[a]Median (range), [b]mean (range), [c]95% confidence interval, [d]median

(mean 45 months) and no pCND (mean 60 months) groups. Zuniga and Sanabria [36] also observed no significant difference in 5-year disease-free survival between the two groups ($p = 0.72$).

Other complications

Chyle leakage, hematoma, wound infection, and hemorrhage were reported in two, three, six, and five studies, respectively. No significant difference in the occurrence of chyle leakage [27, 32], hematoma [5, 10, 32], wound infection [5, 24, 26–28, 35], or hemorrhage [24, 26–28, 35] was observed between the pCND and no pCND groups.

Discussion

The use of pCND in patients with cN0 PTC has always been a topic of debate. The recommendations of guidelines varied and so did the results of many previous studies investigating this topic. The results of this meta-analysis indicate that patients who underwent pCND had a significantly lower chance of experiencing LRR than did those who did not undergo pCND. Nevertheless, these patients had a significantly higher incidence rate of transient RLN injury, transient hypocalcemia, and permanent hypocalcemia.

Our results clearly reveal the advantages and disadvantages of pCND. Considering that patients with PTC generally have an excellent prognosis, a 2% decrease in LRR may not outweigh the risk of adverse events that may significantly affect the quality of life of patients. Only three studies [9, 21, 36] included in this meta-analysis investigated the overall or disease-free survival, and the results of these studies varied. Future studies should investigate the long-term outcomes of pCND to provide valuable insights. As of now, it is crucial for surgeons to evaluate the benefits and consequences of pCND for their patients in order to guarantee optimal outcomes.

The other circumstance that might influence the disease-free survival was ^{131}I ablation therapy. Patients who underwent CND should have received postoperative RAI more frequently than their counterparts who did not undergo CND [36]. Moreover, pCND as an adjunct to total thyroidectomy identified positive nodes in over 30% of patients with PTC. The discovery of positive nodes was associated with higher doses of RAI for postoperative ablation, and there is a trend toward decreased recurrence in patients undergoing pCND [4].

Although complication rates were consistently higher in the pCND group, many retrospective studies investigating this topic have included patient data from as long as 20 years ago. Technological advancements in surgical instruments, such as intraoperative nerve monitoring, have been shown to significantly reduce transient RLN injury [37]. If complication rates, associated with the use of the most advanced instruments, are lower than those reported in retrospective studies, the downside of pCND could be a nonfactor.

Seven studies [9, 18, 23, 24, 27, 31, 32] specifically reported the outcomes of patients who underwent bilateral pCND, and six studies [18, 23, 28, 30, 31, 33] reported the outcomes of patients who underwent ipsilateral pCND. When the subgroups were compared, no significant difference in LRR or any side effects, including transient and permanent RLN injury and hypocalcemia, was observed. Bilateral or ipsilateral pCND did not seem to affect the outcome or complication rates.

Considerable heterogeneity was observed across the studies included in our analysis because of various clinical factors. First, several outcome measurements, such as transient and permanent RLN injury and hypocalcemia, were not standardized. Second, the surgical methods and surgical equipment used in different studies were

Table 2 Methodological quality assessment of included studies

Study	Study design	Preintervention		During intervention	Postintervention				Overall bias
		Bias due to confounding	Selection bias	Bias in classification of interventions	Deviation from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of reported results	
Barczynski et al. [9]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Calò et al. [10]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Calò et al. [18]	Retrospective	Moderate	Moderate	Low	Low	Serious	Low	Serious	Serious
Choi et al. [17]	Retrospective	Moderate	Moderate	Low	Low	Low	Low	Low	Moderate
Costa et al. [21]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Dobrinja et al. [22]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Giordano et al. [23]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Hughes et al. [24]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Hyun et al. [25]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Korkmaz et al. [26]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Kim et al. [27]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Lang et al. [28]	Prospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Lee et al. [29]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Moo et al. [24]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Palestini et al. [30]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Popadich et al. [5]	Prospective	Low	Low	Low	Low	Low	Low	Low	Low
Raffaelli et al. [31]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
So et al. [32]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Sywak et al. [33]	Prospective RCT	Low	Low	Low	Low	Low	Low	Low	Low
Viola et al. [34]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Ywata de Carvalho et al. [11]	Retrospective	Low	Moderate	Low	Low	Low	Low	Low	Moderate
Zhang et al. [35]	Retrospective	Low	Low	Low	Low	Low	Low	Low	Low
Zuniga and Sanabria [36]	Prospective	Low	Low	Low	Low	Low	Low	Low	Low

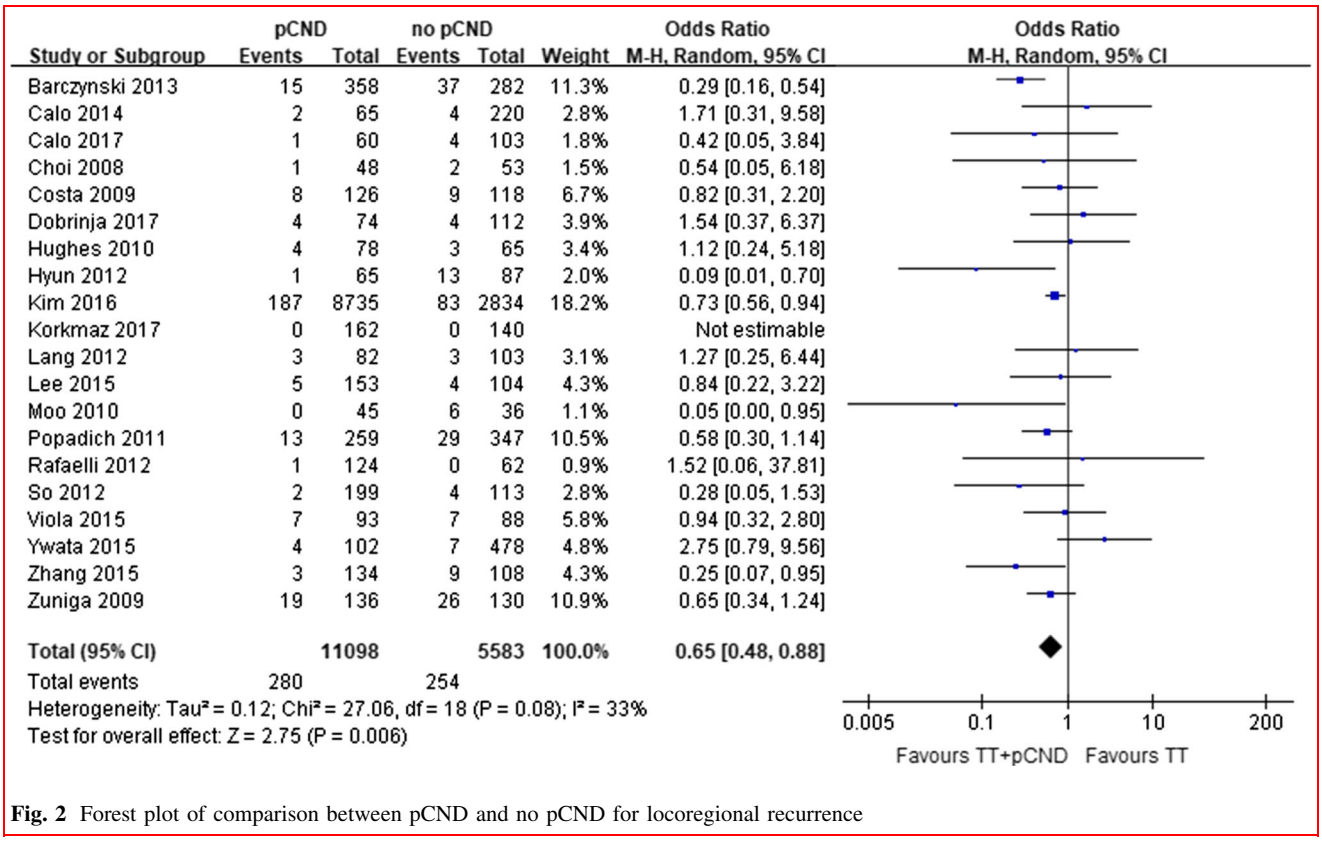

Fig. 2 Forest plot of comparison between pCND and no pCND for locoregional recurrence

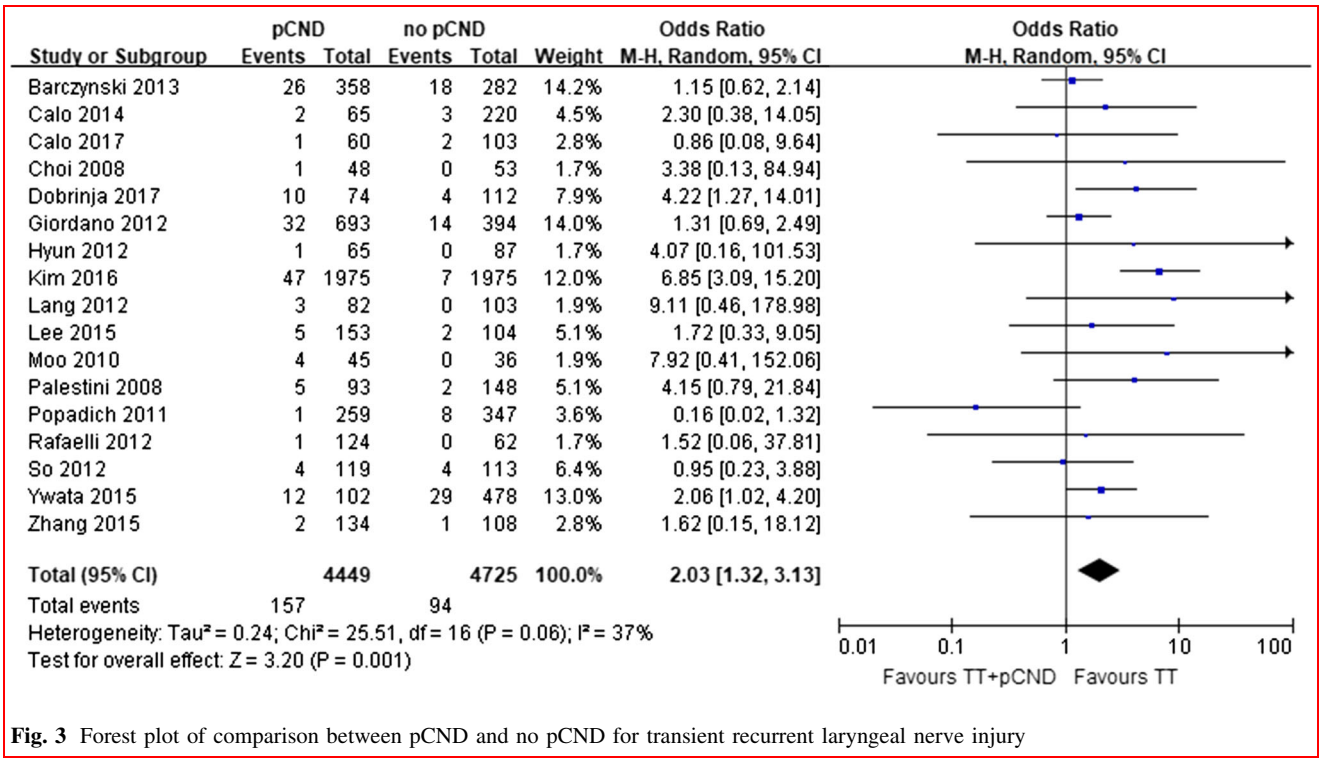

Fig. 3 Forest plot of comparison between pCND and no pCND for transient recurrent laryngeal nerve injury

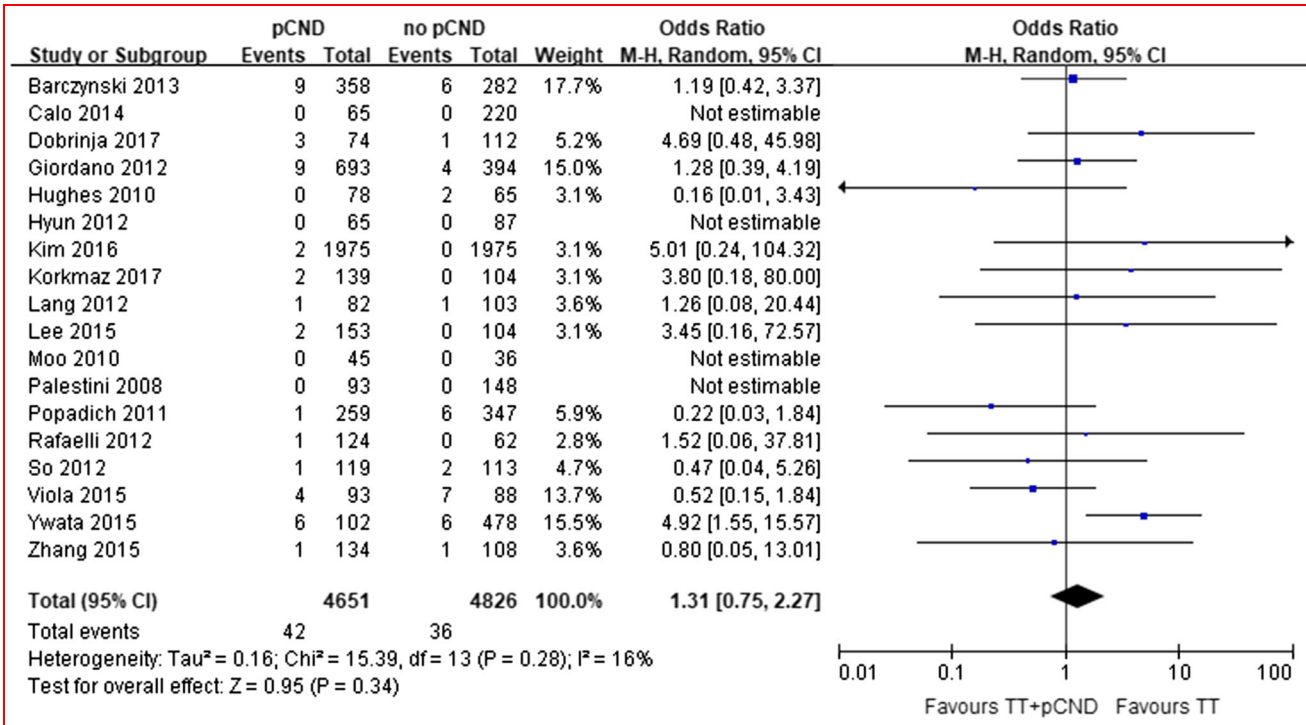

Fig. 4 Forest plot of comparison between pCND and no pCND for permanent recurrent laryngeal nerve injury

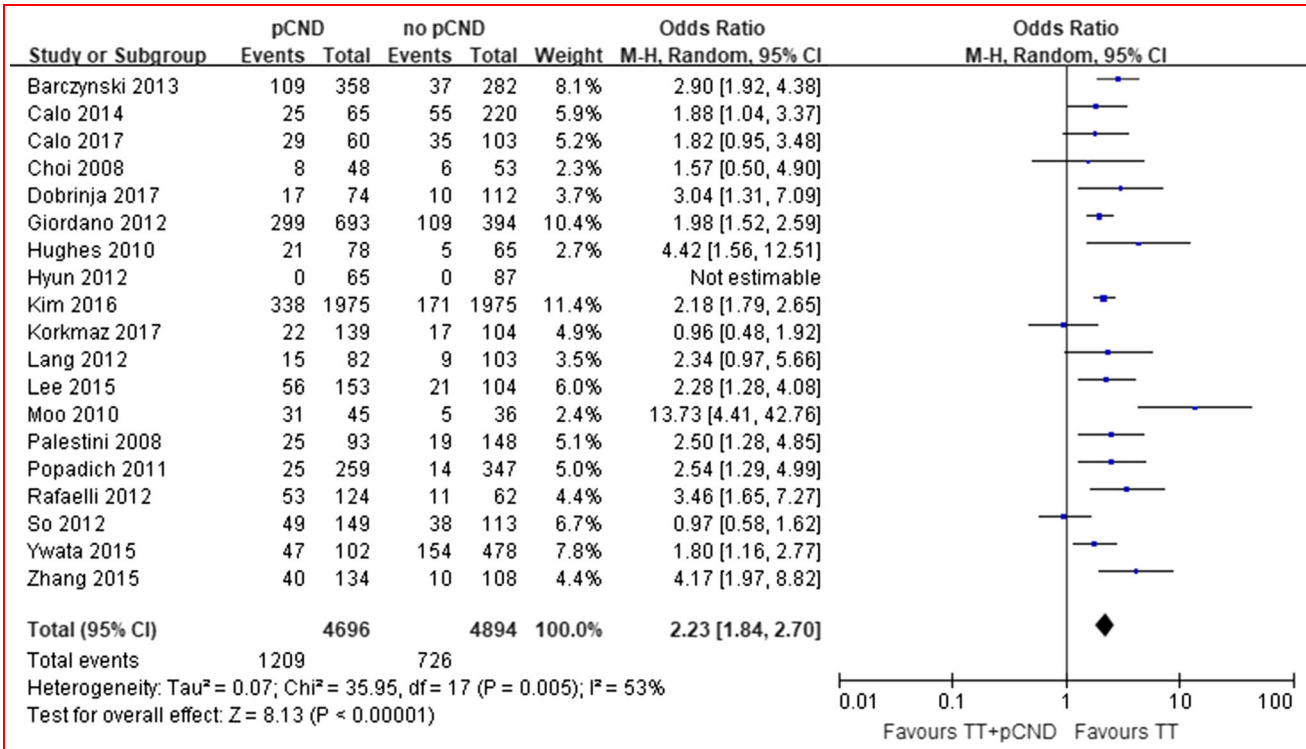

Fig. 5 Forest plot of comparison between pCND and no pCND for transient hypocalcemia

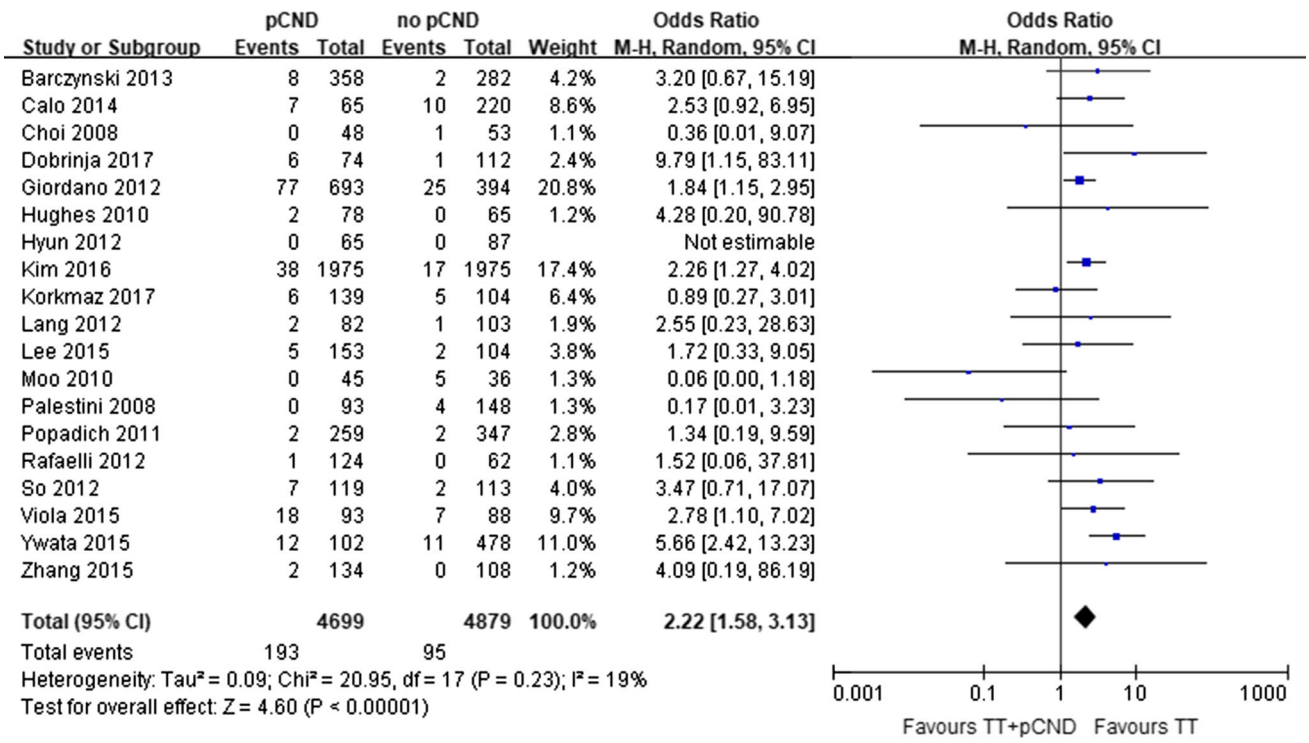

Fig. 6 Forest plot of comparison between pCND and no pCND for permanent hypocalcemia

inconsistent. Differences in the experience level of surgeons might have also contributed to the data heterogeneity.

The current meta-analysis has several limitations. First, the majority of the included studies were retrospective studies. The level of evidence of these studies limits the power of the meta-analysis. Second, the follow-up period of the studies was relatively short. Overall survival and disease-free survival rates were seldom reported; thus, evaluating whether the benefits of pCND outweigh potential costs was difficult. In addition, selection bias cannot be ruled out in many of the included studies. Finally, many of the studies allowed surgeons to decide whether to perform pCND, which possibly introduced selection bias.

Conclusion

The results of our meta-analysis reveal that pCND significantly reduces LRR in patients with cN0 PTC. However, pCND is also associated with significantly higher chances of transient RLN injury, transient hypocalcemia, and permanent hypocalcemia. In addition to TT, pCND provides significant benefits for patients with cN0 PTC but is accompanied by several adverse effects. The clinical decision should be made after a shared decision-making process of clinicians and patients.

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Compliance with ethical standards

Conflict of interest The authors have no conflicts of interest or financial ties to disclose.

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